Illustrate the main properties of a lambertian source.

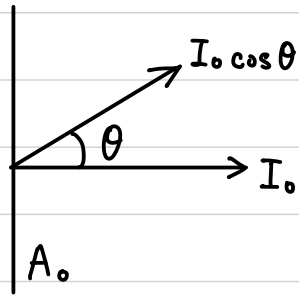
All light sources, either primary or secondary, that appear alike from any observation angle are called lambertian:

$$L(x, y; \theta, \phi) = L(x, y).$$

Especially, when the source is uniform, we have:

$$L(x, y; \theta, \phi) = L = \text{const}.$$

There are some properties of lambertian sources.



1. Lambert's cosine law

The radiant intensity emitted by a small lambertian source is given by:

$$I(\theta, \phi) = \frac{d\Phi}{d\Omega} = \int_{A_0} L \cos\theta \, dA = L A_0 \cos\theta = I_0 \cos\theta,$$

where θ is the angle made by the normal of the lambertian source and the direction of the observer's line of sight.

2. The radiant flux by a lambertian source

The radiant flux emitted by the lambertian source of small area A_0 can be calculated by integrating $I(\theta, \phi)$ over the 2π hemisphere:

$$\Phi = \int_{2\pi} I_0 \cos\theta \, d\Omega = \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} I_0 \cos\theta \sin\theta \, d\theta = \pi I_0.$$

Therefore we simply have:

$$\Phi = \pi L A_0, \quad I_0 = L A_0, \quad M = \pi L.$$

Define the responsivity and the Noise Equivalent Power (NEP) of a radiation detector.

The responsivity is defined as:

$$R_I(\lambda) = \frac{I_{out}[A]}{P_{in}[W]} \quad , \quad R_V(\lambda) = \frac{V_{out}[V]}{P_{in}[W]}$$

and this parameter typically depends on the wavelen th of the radiation field.

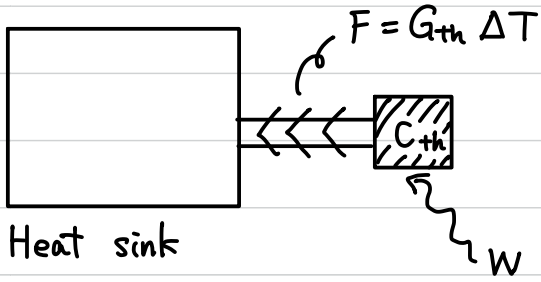
The NEP is a measure of sensitivity and defined as: the optical power in sinusoidal regime that would produce a modulated signal in the noise free detector equal to the r.m.s noise measured from the detector without any input.

Therefore, this can be formulated as follows:

$$NEP_\lambda = \frac{S_{rms}}{R_\lambda} = \frac{N_{rms}}{R_\lambda}.$$

Synthetically describe the main features of thermal detectors.
Illustrate the operation principle of pyroelectric detectors.

In thermal detectors, an absorption material is connected through a thermal link to a massive heat sink. The temperture of the thermal sink is kept at the equilibrium temperature T_{eq}.



When a radiation deliver a power W onto the absorption material, it increases the temperature of the absorption material by ΔT, which determines the thermal flow $F = G_{th} \Delta T$ to the thermal sink. The response of the thermal detector stems from the conversion of the temperature difference ΔT into an electrical signal.

When the optical stimulus is given by a sinusoide superimposed by a continuous wave (CW) component:

$$P(t) = W_0 + W_m \sin(2\pi f t),$$

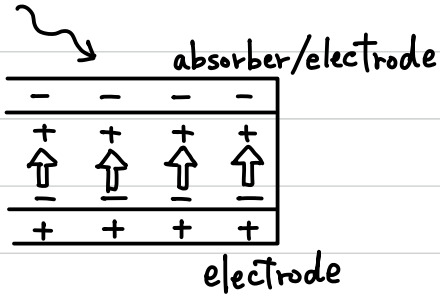
the temperature difference $\Delta T(t)$ after a transient is written as:

$$\Delta T(t) = \frac{W_0}{G_{th}} + \frac{W_m}{G_{th}} \frac{\sin(2\pi f t)}{\sqrt{1 + 4\pi^2 f^2 \tau_{th}^2}}.$$

From this equation, the cutoff frequency f_c is given by:

$$f_c = \frac{1}{2\pi \tau_{th}}.$$

In pyroelectric detectors, a thin slab of ferroelectric material suitably poled and oriented is sandwitched between electrodes, which makes the electrodes equivalent to a capacitor that store a charge Q. The increase in temperature, upon IR radiation, induces the variation of the polarization charge, resulting in a current $i(t)$:



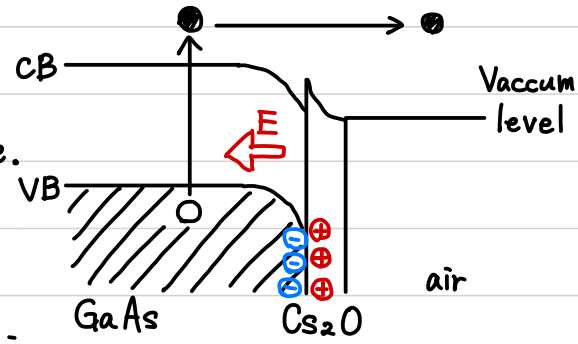
$$i(t) = \frac{dQ(t)}{dt} = A\frac{d\sigma(T(t))}{dt} = A\underbrace{\frac{d\sigma}{dT}}_{p\ (\text{pyroelectric constant})}\frac{dT}{dt} = Ap\frac{dT}{dt}.$$

When the pyroelectric detector receive a sinusoidally modulated optical power: $P(t) = W_0 + W_m \sin(2\pi f t)$, the time derivative of its temperature is described by:

$$\frac{dT(t)}{dt} = \frac{d\,\Delta T(t)}{dt} = \frac{W_m}{G_{th}}\cos(2\pi f t)\frac{2\pi f}{\sqrt{1+4\pi^2 f^2 \tau_{th}^2}}$$

Using a band model, explain the main properties of a Negative Electron Affinity (NEA) photocathode and compare its features with those of a standard semiconductor photocathode.

NEA photocathodes are made of a p-doped semiconductor such as GaAs with a layer of Cs or Cs_2O on top of the GaAs layer. Cesium is an alkali metal prone to loose electron, which leads to the behaviour as an n-type material. Thus, an electric field is produced inward, the vacuum level being pulled downward and the electron affinity becoming negative. Because of this property, electrons can escape even after thermalisation.



The sensitivity of NEA photocathode is extensive and the quantum efficiency is higher compared to standard semiconductor photocathodes.

The operating principles and the main features of photomultiplier tubes

Light from the window induces photoelectrons from the photocathode. These electrons are accelerated and strike the first dynode, resulting in electron multplication by secondary emission. This secondary emission is repeated at each subsequent dynodes and electrons are collected by the anode.

Photomultiplier tubes have a high sensitivity and gain. However, they are fragile, require a high supply voltage and are sensitive to magnetic fields. In addition, spurious pulses with small amplitudes, called afterpulses, may be observed following the signal pulses.

The main sources of noise in photomultiplier tubes

1. The variation of gain in the first dynode

Since the relative variance (RV) of the gain G of photomultiplier:

$$RV_G = \frac{\sigma_G^2}{\langle G \rangle^2}$$

is expressed by:

$$RV_G = RV_1 + \frac{RV_2}{\delta_1} + \frac{RV_3}{\delta_1 \delta} + \cdots + \frac{RV_n}{\delta_1 \delta^{n-2}}$$

By assuming the multiplication processes obey the Poisson distribution, we have

$$RV_G = \frac{1}{\delta_1} + \frac{1}{\delta_1 \delta} + \cdots + \frac{1}{\delta_1 \delta^{n-1}},$$

which indicates the first dynode affects RV_G the most.

2. The shot noise and thermal noise

The shot noise is expressed as:

$$i_{shot}^{RMS} = \sqrt{2(eG)(G\langle I_s \rangle)F\Delta f} = \sqrt{2eG^2\langle I_s \rangle F\Delta f}$$

where G is the gain and F is the noise factor.

Therefore, the total noise is

$$i_n^{RMS} = \sqrt{\left(2eG^2\langle I_s \rangle F + \frac{4kT}{R_L}\right)\Delta f},$$

The operating principles and the main features of avalanche photodiodes

When a reverse bias is applied to a pn junction, primary electrons and holes generated within the depletion layer can acquire enough kinetic energy from the electric field to generate secondary couples of e-h by impact ionisation. A sequence of impact ionisation events can lead to an avalanche process that results in a multiplication factor M up to 100 or more. It increases the responsivity and reduce the NEP.

The main noise sources in avalanche photodiodes

The main noise sources in avalanche photodiodes are the shot noise by signal and dark current and thermal noise. It is formulated as follows:

$$\sqrt{\langle i_n^2 \rangle} = \sqrt{\left[2eM^2(\langle i_s \rangle + \langle i_d \rangle) F_A + \frac{4kT}{R_L}\right] \Delta f},$$

where the shot noise is affected by the noise factor F_A and the maltiplication factor M.

Compare photomultipliers with photodiodes in low light conditions.

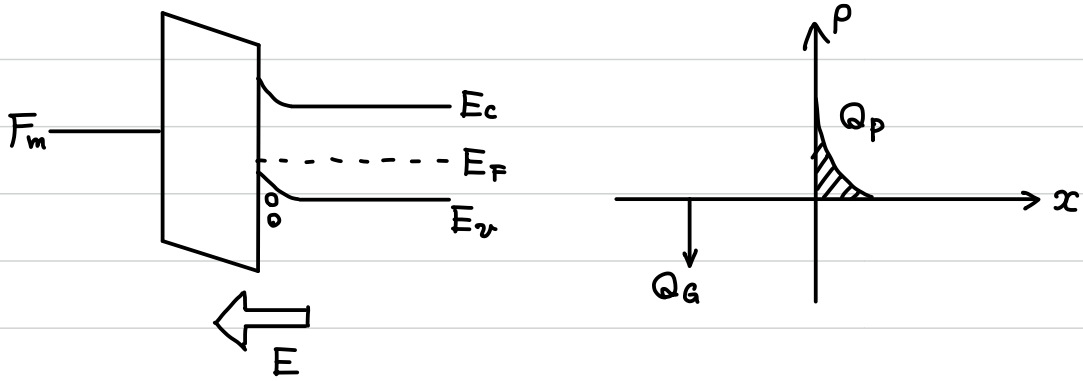
Compare avalanche and PIN photodiode in terms of SNR

Illustrate the behaviour of the MOS capacitor in the different regimes.

- Accumulation regime

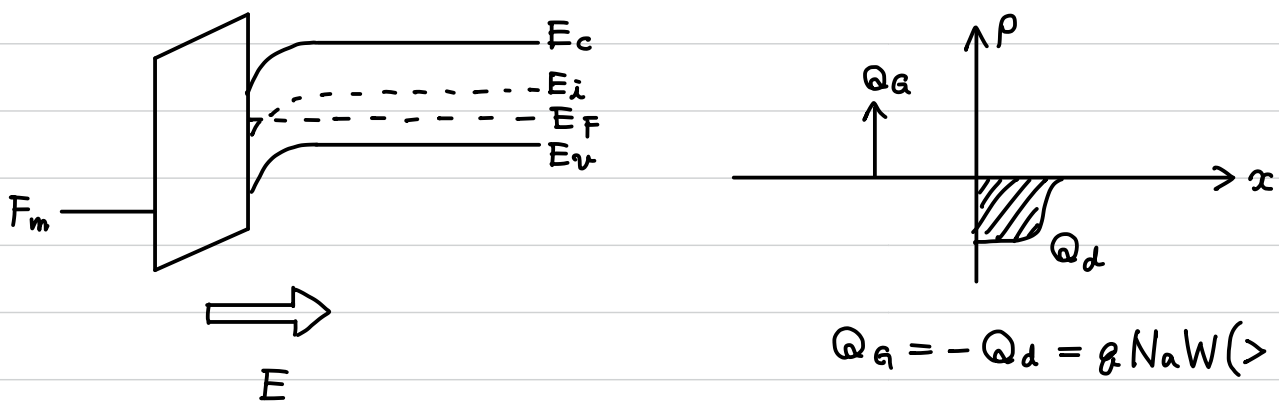
When we apply a negtive electric field, it is established in the semiconductor and the insulator because the metal cannot sustain any electric field. This brings about the acculmulation of the positive charge.

$V_G < 0$



- Depletion regime

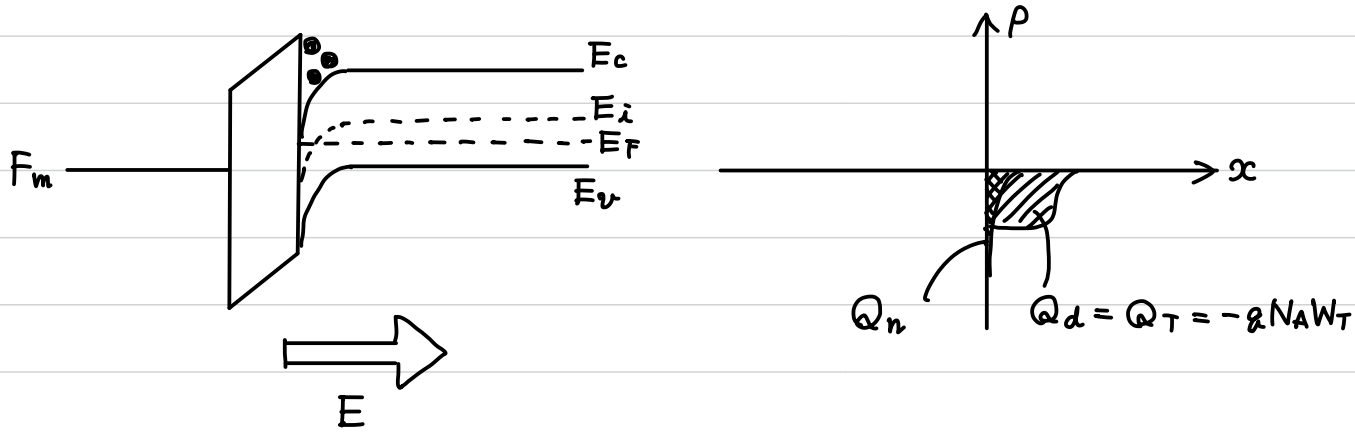
If a positive voltage smaller than the threshold voltage V_T is applied, the bands of semiconductor are bend downward. Therefore, the holes are pushed away from the interfce, which forms a depletion region with a negative charge qN_aW.



$$Q_G = -Q_d = qN_aW (> 0)$$

- Inversion regime

When a strong positive gate voltage ($0 < V_T < V_G$) is applied, not only the depletion region but also the electron charge concentration are produced. The threshold voltage V_T is defined as the voltage where $n_{interface} = p_{bulk} = N_a$.



- Deep depletion regime

The deep depletion regime is attained by sweeping V_G rapidly from $V_G < 0$ to $V_G > V_T$. In this case, holes, being the majority carrier, move in order to reach the appropriate density in all the regions of the semiconductor. This is because electrons in p-region cannot quickly create a negative charge corresponding to the electron density obtained at the interface in the inversion regime.

Synthetically illustrate the main noise sources in scientific grade CCD detectors

There are four main sources.

1. Fixed pattern noise (N_f)

This noise is caused by variations in sensitivity between pixels, These variations may be due to the quantum efficiency differences, non-uniformaties in the aperture area and in the thickness and electronics which are devoted to image readout.

2. Shot noise of the signal (N_s)

This noise is generated by statistical variation in the photon flux incident on a CCD. Shot noise is expressed by the standard deviation of the number of photons collected per pixel during the exposure time. N_s can be formulated as : $N_s = \sqrt{S}$, where S is the signal.

3. Dark shot noise (n_d)

This noise is caused by dark current. The dark current largely depends on the quality of silicon and can be effectively diminished with cooling.

4. Readout noise (N_r)

This noise is mainly due to electrical fluctuations from thermal noise brought about by the MOS-FET used as the amplifier in the CCD output stage. It also comes from the readout circuit of the CCD. The readout noise is the dominant noise source in high performance imaging systems.